

Course: Fundamentals of Generative AI and Large Language Models: Theory and Practice

Week 4: Advanced Variational Autoencoders & Practical Applications

Week 4 Outline

Advanced VAEs & Practical Applications

- Entanglement, Disentanglement Latent Space, and beta-VAE (β -VAE)
- Conditional VAE (cVAE)
- Latent space interpolation

Hands-on Implementation of Variational Autoencoders, cVAE, and β -VAE

Today's Session Overview

- Entanglement + Disentanglement Latent Space

- beta-VAE

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Entangled Latent Space

Consider an image of size: 28×28

Then the input has 784 pixel values:

$$x = [\text{pixel}_1, \text{pixel}_2, \text{pixel}_3, \dots, \text{pixel}_{784}]$$

These pixels may represent many details, but the important variations may be fewer, for example:

thickness, slant, rotation, position

In a VAE, the encoder maps the input to: $\mu_\phi(x)$ and $\sigma_\phi(x)$

and the latent vector is obtained as: $\mathbf{z} = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$

Suppose the latent size is 4, the encoder compresses the image information into four latent dimensions:

$$\mathbf{z} = [z_1, z_2, z_3, z_4]$$

The hidden layer has:

$$\mathbf{h} = [h_1, \dots, h_8]$$

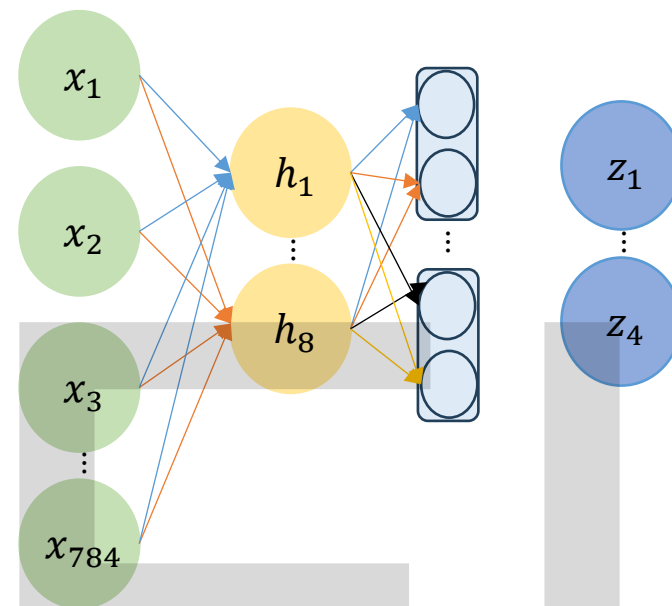
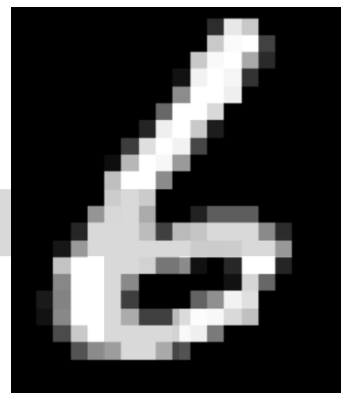
$$h_1 = \text{combination of } x_1, x_2, x_3, \dots, x_{784}$$

$$h_8 = \text{combination of } x_1, x_2, x_3, \dots, x_{784}$$

$\therefore, z_1, z_2, z_3, z_4 = \text{mixed combinations of thickness, slant, rotation, position}$

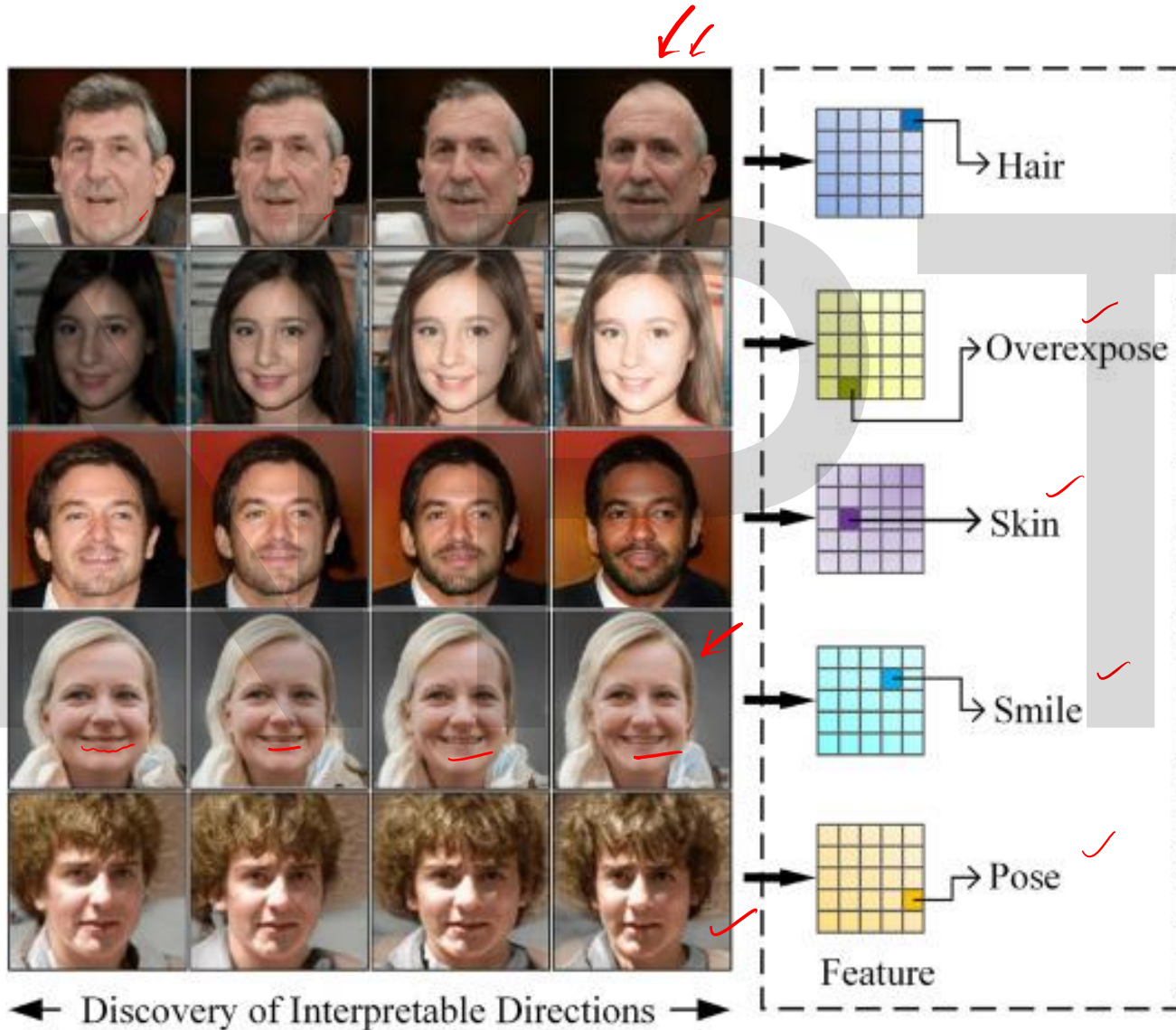
Each latent variable may contain different amounts of these properties.

This is called an **entangled latent representation**.



In an entangled latent representation, changing one latent variable may change multiple properties together.

Consider this



For example:

$z_1 \rightarrow \text{Hair}$ ✓
 $z_2 \rightarrow \text{Lighting/Overexpose}$ ✓
 $z_3 \rightarrow \text{skin}$
 $z_4 \rightarrow \text{smile}$
 $z_5 \rightarrow \text{pose}$ ✓

Each latent variable mainly controls one factor of variation.

Therefore, if we change only z_1 , mainly the hair-related feature should change. Other factors such as lighting, skin tone, smile, and pose should remain mostly unchanged.

Thus, **by manipulating a specific latent variable, the model can generate controlled variations in the image.**

Disentangled Latent Space

A disentangled latent space aims to learn a structured representation where each latent dimension mainly captures one meaningful factor of variation.

Suppose we choose latent size: 5

$$z = [z_1, z_2, z_3, z_4, z_5]$$

For face images, the meaningful factors may be:

hair, lighting, skin tone, smile, pose

$$z_1 \approx \text{hair}, z_2 \approx \text{lighting}, z_3 \approx \text{skin tone}, z_4 \approx \text{smile}, z_5 \approx \text{pose}$$

Changing one latent variable changes mainly one factor:

$z_1 \uparrow \Rightarrow$ hair-related variation changes mainly

$z_2 \uparrow \Rightarrow$ lighting changes mainly

$z_3 \uparrow \Rightarrow$ skin tone changes mainly

$z_4 \uparrow \Rightarrow$ smile changes mainly

$z_5 \uparrow \Rightarrow$ pose changes mainly



If we change one latent variable, mainly one property of the generated image should change, while other properties remain mostly unchanged.

This makes the latent space **interpretable** and **controllable**. This is a **disentangled latent representation**.

β - VAE: Encouraging Disentanglement

A Standard VAE loss is:

$$L(x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] + D_{KL}(q_\phi(z|x) \parallel p(z))$$

The KL term is still there in Standard VAE, but with weight 1, the model may still accept some mixed form across many latent variables if it helps reconstruction. Therefore, even after VAE training, the latent variables may still contain mixed information (**Entangled latent representation**). *Does not guarantee disentanglement*

β - VAE modifies this by adding a weight β to the KL divergence term:

$$L_{\beta\text{-VAE}}(x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] + \beta D_{KL}(q_\phi(z|x) \parallel p(z))$$

The reconstruction loss: Encourages the decoder to reconstruct the input, so that

$$\hat{x} \approx x$$

Store enough information in z so that the decoder can reconstruct the image well.

The KL loss: Encourages the latent distribution $q_\phi(z|x)$ close to the standard normal prior $p(z)$.

Do not store information in an arbitrary way.

So the model is balancing two things:

Good reconstruction vs. Regular latent space

In β - VAE : $\beta > 1$

The KL term receives more importance. The larger KL weight puts stronger pressure on the encoder distribution $q_\phi(z|x)$ to stay close to the prior $p(z) = \mathcal{N}(0, I)$

So it encourages:

$$\mu(x) \approx 0 \quad \sigma^2(x) \approx 1$$

This restricts the model from freely storing mixed information in all latent variables.

$$L_{\beta\text{-VAE}}(x) = -\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] + \beta D_{KL}(q_{\phi}(z|x) \parallel p(z))$$

Therefore:

β -VAE encourages disentangled latent representations by applying stronger regularization.

But:

It does not guarantee perfect disentanglement.

Also, if β is too large:

Reconstruction quality may decrease

because the model focuses too much on keeping $q_{\phi}(z|x)$ close to $\mathcal{N}(0, I)$, and may not store enough details for accurate reconstruction.

Latent Size = 2

If the latent size is 2, the encoder outputs:

$$q_{\phi}(z_1 | x) = \mathcal{N}(\mu_1, \sigma_1^2)$$

$$q_{\phi}(z_2 | x) = \mathcal{N}(\mu_2, \sigma_2^2)$$

Using the reparameterization trick:

$$z_1 = \mu_1 + \sigma_1 \epsilon_1, \quad z_2 = \mu_2 + \sigma_2 \epsilon_2$$

where

$$\epsilon_1, \epsilon_2 \sim \mathcal{N}(0,1)$$

In an ideal disentangled case:

$z_1 \rightarrow$ *mainly thickness*

$z_2 \rightarrow$ *mainly slant*

So, each latent dimension is expected to carry one meaningful factor of variation.

If the latent representation is disentangled, then the distribution for z_1 , especially μ_1 , mainly encodes thickness.

Similarly, z_2 mainly carries slant information.

Suppose:

$$\mu_1 = 0.4$$

$$\mu_2 = 0.2$$

For simplicity, assume:

$$\sigma_1^2 = 1, \quad \sigma_2^2 = 1$$

Now calculate the KL cost.

For one latent dimension:

$$D_{KL} = \frac{1}{2}(\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$$

Since:

$$\sigma_j^2 = 1$$

we get:

$$D_{KL} = \frac{1}{2}(\mu_j^2 + 1 - \log 1 - 1)$$

$$D_{KL} = \frac{1}{2}\mu_j^2$$

For two latent dimensions:

$$D_{KL} = \frac{1}{2}(\mu_1^2 + \mu_2^2)$$

Substitute:

$$\mu_1 = 0.4, \quad \mu_2 = 0.2$$

$$D_{KL} = \frac{1}{2}(0.4^2 + 0.2^2)$$

$$D_{KL} = \frac{1}{2}(0.16 + 0.04)$$

$$D_{KL} = \frac{1}{2}(0.20)$$

$$D_{KL} = 0.10$$

The distribution is close to the prior $\mu = [0,0]$ and more regularized. KL value is small.

In the entangled case, we say:

$z_1 \rightarrow$ mainly carries mixed thickness and slant information

$z_2 \rightarrow$ also carries mixed thickness and slant information

For example, assume :

$\mu_1 =$ contains thickness + slant

$\mu_2 =$ also contains mixed information

Suppose:

$$\mu_1 = 0.8$$

$$\mu_2 = 0.6$$

For simplicity, assume:

$$\sigma_1^2 = 1, \quad \sigma_2^2 = 1$$

Now calculate the KL cost.

For one latent dimension:

$$D_{KL} = \frac{1}{2} (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$$

Since:

$$\sigma_j^2 = 1$$

we get:

$$D_{KL} = \frac{1}{2} (\mu_j^2 + 1 - \log 1 - 1)$$

$$D_{KL} = \frac{1}{2} \mu_j^2$$

For two latent dimensions:

$$D_{KL} = \frac{1}{2} (\mu_1^2 + \mu_2^2)$$

Substitute:

$$\mu_1 = 0.8, \quad \mu_2 = 0.6$$

$$D_{KL} = \frac{1}{2} (0.8^2 + 0.6^2)$$

$$D_{KL} = \frac{1}{2} (0.64 + 0.36)$$

$$D_{KL} = \frac{1}{2} (1.00)$$

$$D_{KL} = 0.50$$

The distribution is farther from the prior and the encoder is using more latent capacity to store information.

In the entangled case, we say:

$z_1 \rightarrow$ mainly carries mixed thickness and slant information

$z_2 \rightarrow$ also carries mixed thickness and slant information

For example, assume :

$\mu_1 =$ contains thickness + slant

$\mu_2 =$ also contains mixed information

Suppose:

$$\mu_1 = 1.5$$

$$\mu_2 = 0.5$$

For simplicity, assume:

$$\sigma_1^2 = 1, \quad \sigma_2^2 = 1$$

Now calculate the KL cost.

For one latent dimension:

$$D_{KL} = \frac{1}{2} (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$$

Since:

$$\sigma_j^2 = 1$$

we get:

$$D_{KL} = \frac{1}{2} (\mu_j^2 + 1 - \log 1 - 1)$$

$$D_{KL} = \frac{1}{2} \mu_j^2$$

For two latent dimensions:

$$D_{KL} = \frac{1}{2} (\mu_1^2 + \mu_2^2)$$

Substitute:

$$\mu_1 = 1.5, \quad \mu_2 = 0.5$$

$$D_{KL} = \frac{1}{2} (1.5^2 + 0.5^2)$$

$$D_{KL} = \frac{1}{2} (2.25 + 0.25)$$

$$D_{KL} = \frac{1}{2} (2.5)$$

$$D_{KL} = 1.25$$

The distribution is farther from the prior and the encoder is using more latent capacity to store information.

If the image has **many important factors of variation**, but the latent space has **fewer latent variables**

For example, suppose the image has 6 important factors:

hair, *lighting*, skin tone, smile, pose, background

but the latent size is only 3:

Then there are two possibilities:

Case 1: Entanglement happens

One latent variable may capture a mixture of factors.

Since the model is storing more information in the latent variables, reconstruction may be better.

So:

Reconstruction Loss may be lower.

$$z = [z_1, z_2, z_3]$$

But because the latent variables are carrying more information, the encoder distribution may move farther from the prior $N(0, I)$.

So:

$$KL(q_\phi(z | x) \parallel p(z)) \text{ may increase}$$

So, the β value is usually **low or not strong enough** to enforce disentanglement.

If the image has **many important factors of variation**, but the latent space has **fewer latent variables**

For example, suppose the image has 6 important factors:

hair, *lighting*, skin tone, smile, pose, background

but the latent size is only 3:

$$z = [z_1, z_2, z_3]$$

Then there are two possibilities:

Case 2: We encourage disentanglement with fewer latent variables: Each latent variable to capture only one factor.

Then the model may capture only a few dominant factors, while the remaining factors may be ignored or weakly represented.

Therefore, the decoder receives incomplete information, reconstruction quality decreases, and reconstruction loss increases.

Because we are encouraging disentanglement,

$$\beta > 1$$

A higher β gives more importance to the KL term.

This pushes the encoder distribution $q_\phi(z | x)$ to stay closer to the prior:

$$p(z) = N(0, I)$$

So:

$$KL(q_\phi(z | x) \parallel p(z)) \text{ may decrease } \downarrow$$

Possible solution

Solution 1:

Increase latent size:

$$z = [z_1, z_2, z_3] \rightarrow z = [z_1, z_2, z_3, z_4, z_5, z_6]$$

This gives the model more capacity to capture more meaningful factors separately.

More latent capacity $\Rightarrow$ more information captured $\Rightarrow$ better reconstruction $\Rightarrow$ lower reconstruction loss

Solution 2: Tune the β value

If β is too high:

$\beta \uparrow \Rightarrow$ strong KL pressure $\Rightarrow$ less information stored in $z \Rightarrow$ reconstruction loss increases

So one solution is to choose a moderate β , not too high.

Moderate $\beta \Rightarrow$ balance between reconstruction and disentanglement

Solution 3: Use β annealing

Instead of using a high β from the beginning, start with a smaller value and gradually increase it.

$$\beta: 1 \rightarrow 2 \rightarrow 4$$

This allows the model to first learn good reconstruction and then gradually improve latent structure. β annealing, β may start from 0 or a very small value only during the initial training stage. At this stage, the KL term has little or no effect, so the model learns reconstruction first. Then β is gradually increased so that the KL term becomes active and starts encouraging disentanglement.

Next Session

➤ Conditional VAE

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Thank you